

5.2 Basis & Dimension Computations

- * calculate bases for subspaces
- * transpose switches columns/rows

$$\begin{aligned} * \text{Col } A &= \text{Row } A^T \\ \text{Col } A^T &= \text{Row } A \end{aligned}$$

$$* V = \text{Col } A \Rightarrow V^\perp = (\text{Col } A)^\perp = (\text{Row } A^T)^\perp = \text{Null } A^T$$

$$\text{Row}^\perp = \text{Null}$$

$$\hookrightarrow RA^T = \text{Null}$$

① Bases for Col A & Row A

- * Col A basis = pivot columns of A
- * Row A basis = non-zero rows of A_{red}

$$\Rightarrow \text{Rank } A = \dim \text{Col } A = \dim \text{Row } A = \# \text{ pivots in } A_{\text{red}}$$

② Basis for Null A

$$* [A|0] \rightarrow [A_{\text{red}}|0] \rightarrow x = t_1 x_1 + \dots + t_k x_k$$

- * free parameters (per free column in A_{red}) $\rightarrow$ basis $\{x_1, \dots, x_k\}$

$$\Rightarrow \dim \text{Null } A = \# \text{ of free columns}$$

③ Rank Theorem

$$* T_A: \mathbb{R}^n \rightarrow \mathbb{R}^m, A - m \times n$$

$$* \dim \text{Col } A + \dim \text{Null } A = \dim \mathbb{R}^n = n$$

$$* \# \text{ pivot col} + \# \text{ free col} = n$$

$\left\{ \begin{array}{l} \text{Col } A \\ \text{Null } A \end{array} \right\}$ & Null A complementary

④ Basis for $V^\perp$

$$* V^\perp = (\text{Col } A)^\perp = (\text{Row } A^T)^\perp = \text{Null } A^T$$

$$\Rightarrow \text{take transpose \& find basis for Null } A^T$$

⑤ Basis for Col A & Col $A^\perp$

$$\begin{aligned} * \text{Col } A &= \text{Row } A^T \Rightarrow A \rightarrow A^T \rightarrow A^T_{\text{red}} \begin{cases} \text{Col } A = \text{pivot rows as columns} \\ [A^T_{\text{red}}|0] = \text{Col } A^\perp \text{ basis} \end{cases} \\ \text{Col } A^\perp &= \text{Null } A^T \end{aligned}$$

$$* \dim V + \dim V^\perp = \dim \mathbb{R}^n = n \quad (V = \text{Col } A)$$

⑥ Basis for $\mathbb{R}^n$

$$* \text{If } V \cap V^\perp = \{0\} \text{ and } \dim V + \dim V^\perp = n \Rightarrow \text{union is basis for } \mathbb{R}^n$$

$$* \text{To extend basis:}$$

$$\checkmark NA^T = \text{Col } A^\perp$$

$$A \rightarrow A^T \rightarrow \text{basis for Null } A^T = \text{Col } A^\perp \rightarrow \text{union w/ Col } A$$

6.1, 6.2 Matrix Algebra

① Matrices & Linear Transformations

- * sum of transformations

$$* (T_A + T_B)(x) = T_A(x) + T_B(x)$$

$$* T_A + T_B = T_{A+B}$$

$$* \text{operation of transformation} = \text{operation of matrices}$$

- * scale of transformations

$$* (\lambda T_A)(x) = \lambda(T_A(x))$$

$$* \lambda T_A = T_{\lambda A}$$

$$* \text{scaling transformation} = \text{scaling outputs}$$

② Matrix Multiplication

$$* \begin{bmatrix} \text{---} \end{bmatrix} \begin{bmatrix} \text{---} \\ \text{---} \end{bmatrix} = \begin{bmatrix} \text{---} \end{bmatrix} \quad \text{w/ dotless product}$$

- * Facts

$$* AB \neq BA$$

$$* A(B+C) = AB + AC$$

$$* A(\lambda B) = \lambda(AB) = (\lambda A)B$$

$$* (A+B)C = AC + BC$$

$$* A(BC) = (AB)C$$

- * matrix multiplication = composition of linear transformations

$$\begin{array}{ccc} I_m & & A \\ \downarrow \text{row op} & \dots \rightarrow & \downarrow \\ E & & EA \end{array} \quad \begin{array}{l} \text{(multiplication on left by elementary matrix } \\ \text{performs the row operations)} \end{array} \quad [I_n]$$

③ Matrix Transpose w/ operations

- * facts:

$$* (A+B)^T = A^T + B^T$$

$$* (\lambda A)^T = \lambda(A^T)$$

$$* (AB)^T = B^T A^T \quad (\text{order reversed})$$

- * consequence:

$$* T_A: \mathbb{R}^n \rightarrow \mathbb{R}^m$$

$$T_A^T: \mathbb{R}^m \rightarrow \mathbb{R}^n$$

$$\Rightarrow u \cdot (Av) = (A^T u) \cdot v$$

$$\hookrightarrow u \cdot (Av) = u^T (Av) = (u^T A) v = (A^T u)^T v = (A^T u) \cdot v$$

7.1 Invertible Matrices

① Invertibility

- $n \times n$ matrix A , $n \times n$ matrix B invertible if

$$AB = I_n = BA, \quad B = A^{-1}$$

$$AB = I_n \Leftrightarrow BA = I_n$$

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \text{ invertible} \Leftrightarrow ad - bc \neq 0 \quad \& \quad A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

- elementary matrices are invertible w/ elementary matrices

• Properties

$$(A^{-1})^{-1} = A$$

$$(AB)^{-1} = B^{-1}A^{-1}$$

$$I_n^{-1} = I_n$$

$$(A^T)^{-1} = (A^{-1})^T$$

$$(CA)^{-1} = C^{-1}A^{-1}$$

$$A, B \text{ invertible} \not\Rightarrow A+B \text{ invertible}$$

② usage

$$A \text{ invertible} \& \quad Ax = b \Rightarrow x = A^{-1}b$$

$$A \text{ invertible} \Rightarrow Ax = b \text{ has unique solution for all } b \in \mathbb{R}^n \Rightarrow A_{\text{red}} = I_n$$

$$A \text{ invertible} \Leftrightarrow A_{\text{red}} = I_n$$

$$[A | I_n]_{\text{red}} = [I_n | A^{-1}] \quad (\text{b/c } A \text{ invertible} \Leftrightarrow \text{product of elementary matrices})$$

③ Conceptual

$$AB = I_n = BA \Leftrightarrow T_A(T_B(x)) = x = T_B(T_A(x))$$

$$A \text{ invertible w/ inverse } A^{-1} \Leftrightarrow T_A \text{ invertible w/ inverse } T_A^{-1}$$

④ Goldilocks Principle

7.2, 7.3 Triangular & Orthogonal Matrices

① Triangular Matrices

- lower triangular = zero above main diagonal

- upper triangular = zero below main diagonal

- diagonal matrices = upper & lower

• properties

- product of 2 upper triangular = upper triangular

$$A \text{ nxn} \Rightarrow A_{\text{red}} \text{ upper triangular}$$

$$\text{row equivalent to } I_n: \begin{bmatrix} a_1 & & * \\ & \ddots & \\ 0 & & a_n \end{bmatrix} \text{ invertible} \Leftrightarrow a_1 \neq 0, \dots, a_n \neq 0$$

$$A \text{ invertible} \Leftrightarrow \text{columns of } A \text{ form basis for } \mathbb{R}^n$$

② Orthogonal Matrices

- $\{v_1, \dots, v_n\} \subset \mathbb{R}^n$ orthogonal basis if:

$$v_i \cdot v_j = 0 \text{ (pairwise orthogonal)}$$

- Orthonormal basis if:

$$v_i \cdot v_j = 0$$

$$v_i \cdot v_i = 1 \text{ for all } i$$

• Facts

- non-zero pairwise orthogonal vectors are L.I. (form basis)

$$U = [v_1 \dots v_n] \Leftrightarrow \begin{matrix} \text{orthogonal} & \{v_1, \dots, v_n\} \\ & \text{orthonormal basis} \end{matrix}$$

$$U \text{ a } 2 \times 2 \text{ matrix} \Leftrightarrow U \text{ a } 2 \times 2 \text{ rotation/reflection matrix}$$

• General properties

$$U^T U = I_n = U U^T \begin{matrix} \nearrow U \text{ orthogonal} \\ \searrow (Ux) \cdot (Uy) = x \cdot y \text{ (preserve dot product)} \\ \nearrow \|Ux\| = \|x\| \text{ (preserve length)} \end{matrix}$$

- orthogonal operations only change direction, not length *

8.1, 8.2 Determinants

① Determinant Basics

- $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ invertible $\Leftrightarrow ad - bc \neq 0$
- $\det = 0 \Leftrightarrow A$ not invertible
- $\det A \neq 0 \Leftrightarrow \text{Area}(R) \neq 0$
 $\Leftrightarrow$ columns of A a basis for $\mathbb{R}^2$
 $\Leftrightarrow A$ invertible
- $\text{Area}(R) = |\det A|$
- $T_A: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ scales areas by $|\det A|$
- **signs**
 - $\hookrightarrow \det A > 0$: columns oriented in same way standard basis
 - $\hookrightarrow \det A < 0$: columns oriented different way to standard basis
 - $\hookrightarrow \det A = 0$: $\longrightarrow$ lose dim
- $\det(A) + \det(B) \neq \det(A+B)$
- determinants not linear

② Calculating Determinant

- **Permutation Matrix**
 - P is a permutation matrix if columns are the standard basis vectors in any order
 - $\text{sign}(P) = \begin{cases} +1 & \text{even \# of pairs out of order} \\ -1 & \text{odd \# of pairs out of order} \end{cases}$
- **Elementary matrices**
 - $A \xrightarrow{Ex} A'$ w/ $\det A' = -\det A$
 - $A \xrightarrow{Sc} A'$ w/ $\det A' = c \det A$
 - $A \xrightarrow{Sh} A'$ w/ $\det A' = \det A$
 - multiply row ops together until upper triangular
 - $\det AB = \det A \det B$
 - A invertible $\Leftrightarrow \det A \neq 0$ & $\det A^{-1} = 1/\det A$
 - U orthogonal $\Rightarrow \det U = 1$ (for rotations) or $\det U = -1$ (reflections in $\mathbb{R}^2$)
- **Cofactor Expansion**

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix} \rightarrow \begin{bmatrix} \square & \square & \square \\ \square & \square & \square \\ \square & \square & \square \end{bmatrix} \begin{bmatrix} \square & \square & \square \\ \square & \square & \square \\ \square & \square & \square \end{bmatrix} \begin{bmatrix} \square & \square & \square \\ \square & \square & \square \\ \square & \square & \square \end{bmatrix}$$

9.1, 9.2 Linear Coordinate Systems

① Basics

- $V \in V \subset \mathbb{R}^n$ subspace $\Rightarrow V = a_1 v_1 + \dots + a_d v_d$
 $B = \{v_1, \dots, v_d\} \subset V$ basis $\uparrow$
 $a_1, \dots, a_d$ unique for V
- fixing B gives a grid
- $V \in V \subset \mathbb{R}^n$ subspace
 $B = \{v_1, \dots, v_d\} \subset V$ basis $\Rightarrow [V]_B = \begin{bmatrix} a_1 \\ \vdots \\ a_d \end{bmatrix} \in \mathbb{R}^d$
 $V = a_1 v_1 + \dots + a_d v_d$
 $\uparrow$ "B-coordinates of V " weights stacked in a column
- B-coordinate system on V = a linear coordinate system

$$\begin{cases} * P_B = [v_1 \dots v_d] \\ P_B [V]_B = V \end{cases} \leftarrow \text{"B-coords back to regular coords w/ matrix mul"}$$

- $E_n = \{e_1, \dots, e_n\}$ standard basis for $\mathbb{R}^n$
 - $v = \mathbb{R}^n$
 - $[v]_{E_n} = \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} = a_1 e_1 + \dots + a_n e_n$
 $\uparrow$ standard coordinates

② Calculating Coordinates (B-coordinates)

$$\begin{cases} * v \in V \subset \mathbb{R}^n \\ B = \{v_1, \dots, v_d\} \subset V \text{ basis} \end{cases} \Rightarrow [v_1 \dots v_d | v]_{\text{red}} = \begin{bmatrix} I_d & [v]_B \\ 0 & 0 \end{bmatrix}$$

$\uparrow$ $[P_B | v]$

- $\Rightarrow [v]_B = e_1, \dots, [v_d]_B = e_d$
 $\hookrightarrow B$ as standard basis in B-coords
- $[c_1 w_1 + \dots + c_k w_k]_B = c_1 [w_1]_B + \dots + c_k [w_k]_B \rightarrow$ B-coords are linear

③ Change of Basis Matrix

- $V \subset \mathbb{R}^n$ subspace
 $B = \{b_1, \dots, b_d\} \subset V$ bases $\Rightarrow P_{C \leftarrow B} = [[b_1]_C \dots [b_d]_C]$
 $C = \{c_1, \dots, c_d\} \subset V$ basis B in C-coords

$$* [c_1 \dots c_d | b_1 \dots b_d]_{\text{red}} = [P_C | P_B]_{\text{red}} = \begin{bmatrix} I_d & P_{C \leftarrow B} \\ 0 & 0 \end{bmatrix}$$

- $P_{C \leftarrow B} [v]_B = [v]_C \leftarrow$ translate bwn coord systems

$$\begin{array}{ccc} & V & \\ & \swarrow \downarrow \searrow & \\ \mathbb{R}^d & \xrightarrow{P_{C \leftarrow B}} & \mathbb{R}^d \\ [v]_B & & [v]_C \end{array}$$

- $V = \mathbb{R}^n \Rightarrow P_{E_n \leftarrow B} = P_B$ & $P_{C \leftarrow E_n} = P_C^{-1} \Rightarrow P_{C \leftarrow B} = P_C^{-1} P_B$
 $\uparrow$ $B \rightarrow \text{standard} \rightarrow C$

10.1, 10.2 Orthogonal Bases and Coordinates

① Recall

* $V \subseteq \mathbb{R}^n$ subspace

$$B = \{v_1, \dots, v_d\} \subset V \text{ basis} \Rightarrow [v]_B = \begin{bmatrix} a_1 \\ \vdots \\ a_d \end{bmatrix} \in \mathbb{R}^d$$

$$v = a_1 v_1 + \dots + a_d v_d$$

↑ B-coords for v

$$\Rightarrow [P_B | v]_{\text{red}} = \begin{bmatrix} I_d & [v]_B \\ 0 & 0 \end{bmatrix}$$

↑ $[v_1 \dots v_d]$

② Orthogonal/Orthonormal change of Basis

* can skip G.E

$$B = \{v_1, \dots, v_d\} \subset V \text{ orthogonal basis} \Rightarrow [v]_B = \begin{bmatrix} \frac{v \cdot v_1}{v_1 \cdot v_1} \\ \vdots \\ \frac{v \cdot v_d}{v_d \cdot v_d} \end{bmatrix} \in \mathbb{R}^d$$

$$B = \{v_1, \dots, v_d\} \subset V \text{ orthonormal basis} \Rightarrow [v]_B = \begin{bmatrix} v \cdot v_1 \\ \vdots \\ v \cdot v_d \end{bmatrix} = \begin{bmatrix} v_1^T v \\ \vdots \\ v_d^T v \end{bmatrix} = P_B^T v$$

$$B \text{ \& } C \text{ bases} \Rightarrow P_{C \leftarrow B} = P_C^T P_B$$

and C orthonormal

↑ orthonormal matrix if $B \& C$ orthonormal

③ Construct Orthonormal Basis

$$B = \{v_1, \dots, v_d\} \xrightarrow{\text{G.S.}} u = \{u_1, \dots, u_d\} \xrightarrow{\text{normalize}} \hat{u} = \{\hat{u}_1, \dots, \hat{u}_d\}$$

basis for V orthogonal basis V orthonormal basis

• Gram-Schmidt Process

• given $\{v_1, \dots, v_d\} \subset V \rightarrow$ orthogonal basis

$$u_1 = v_1$$

$$u_2 = v_2 - \frac{v_2 \cdot u_1}{u_1 \cdot u_1} u_1$$

• normalize

$$\hat{u}_n = \frac{u_n}{\|u_n\|} = \frac{1}{\sqrt{u_n \cdot u_n}} u_n$$

11.1 Projection Transformation

① Definition

* $V, W \subset \mathbb{R}^n$ subspaces
are complements
written $v \oplus w$

every $x \in \mathbb{R}^n$ can be
written uniquely as
 $x = x_v + x_w$
in V in W

$$\dim V + \dim W = n$$

&

$$V \cap W = \{0\}$$

• Notes

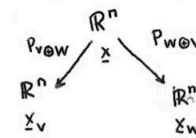
• can project at different angles

• W = how we project

• V = what we projecting to

• $V \subset \mathbb{R}^n$ can have infinitely many complements

* $V, W \subset \mathbb{R}^n$ complements



$$* P_{V \oplus W}(x) + P_{W \oplus V}(x) = x$$

$$\Rightarrow P_{W \oplus V} = x - P_{V \oplus W}(x)$$

one proj gives another

• Algorithm

$$* P_{V \oplus W}(x) = x_v \text{ \& } P_{W \oplus V}(x) = x_w = x - x_v$$

• $\hat{Q}$: Find a basis for $V \{v_1, \dots, v_k\}$ \& $W \{w_1, \dots, w_k\}$.

B/c V, W comp: $\{v_1, \dots, v_k, w_1, \dots, w_k\}$ basis for $\mathbb{R}^n$,

$$x = \underbrace{a_1 v_1 + \dots + a_d v_d}_{x_v} + \underbrace{b_1 w_1 + \dots + b_k w_k}_{x_w}$$

$$\Rightarrow \text{solve: } [v_1 \dots v_d w_1 \dots w_k] x$$

② Special Case - Orthogonal Projection

• $V \subset \mathbb{R}^n$ subspace, $V^\perp = \{x \mid x \cdot v = 0 \forall v \in V\}$

↑ avoids G.E.

• $W = V^\perp$, $P_{V \oplus V^\perp} = P_V$

$$B = \{v_1, \dots, v_d\} \subset V \Rightarrow P_V(x) = \frac{x \cdot v_1}{v_1 \cdot v_1} v_1 + \dots + \frac{x \cdot v_d}{v_d \cdot v_d} v_d$$

orthogonal basis
C.S.

$$P_{V^\perp}(x) = x - P_V(x)$$

$$* \text{ orthonormal basis} \Rightarrow P_V(x) = P_B P_B^T(x) \leftarrow P_B = [v_1 \dots v_d]$$

↑ coef. matrix for projection

• if V = line \& $V^\perp$ = hyperplane, project x to $V^\perp$ \& $P_{V^\perp}(x) = x - P_V(x)$

③ Proj. Abstract Characterization

• $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a projection $\Leftrightarrow A^2 = A$

$$T = P_{V \oplus W} \uparrow$$

↑ orthogonal projection if $A^T = A$

11.2 Geometry of Orthogonal Projections

① Orthogonal Projections

• $\underline{x} \in V \subset \mathbb{R}^n$ a subspace, $\underline{x} = \underline{x}_V + \underline{x}_{V^\perp}$

• $B = \{v_1, \dots, v_d\} \subset V \Rightarrow P_V(\underline{x}) = \frac{\underline{x} \cdot v_1}{v_1 \cdot v_1} v_1 + \dots + \frac{\underline{x} \cdot v_d}{v_d \cdot v_d} v_d$
orthogonal basis

↑ a.s.

$$P_{V^\perp}(\underline{x}) = \underline{x} - P_V(\underline{x})$$

② Minimum Distance

• minimum distance btwn $\underline{x}$ and V :

$$= \|\underline{x} - P_V(\underline{x})\| = \|\underline{P}_{V^\perp}(\underline{x})\| \leftarrow \leq \|\underline{x} - \underline{y}\|, \underline{y} = \text{closest}$$

③ Least-Squares

• If $A\underline{x} = \underline{b}$ inconsistent ($\underline{b} \notin \text{Col}(A)$), best approximate solution

• least-squares solution = solution to $A\underline{x} = P_{\text{Col}(A)}(\underline{b})$ = solution to $A^T A \underline{x} = A^T \underline{b}$
to $A\underline{x} = \underline{b}$

• If $A\underline{x} = \underline{b}$ consistent $\Rightarrow \underline{b} = P_{\text{Col}(A)}(\underline{b}) \Rightarrow$ Least squares sol = ordinary sol.

• Method 1: find $P_{\text{Col}(A)}(\underline{b})$ & solve $A\underline{x} = P_{\text{Col}(A)}(\underline{b})$

↳ given A , do a.s.

↳ O.E. SLE

$$\|P_{\text{Col}(A)}(\underline{b}) - \underline{b}\| = \text{length}$$

④ Normal Equations

• Fact: $A\underline{x} = P_{\text{Col}(A)}(\underline{b}) \Leftrightarrow A^T A \underline{x} = A^T \underline{b}$

• Method 2: $[A^T A | A^T \underline{b}]$
↑ normal Eq. of $A\underline{x} = \underline{b}$
& solve w/o $P_{\text{Col}(A)}(\underline{b})$

$$\uparrow \underline{b} - A\underline{x} = P_{\text{Col}(A)^\perp}(\underline{b})$$

⑤ Application

• $A = [v_1 \dots v_d]$

$\{v_1, \dots, v_d\}$ basis $V = \text{Col } A$

$$\Rightarrow P_V(\underline{v}) = \underline{A(A^T A)^{-1} A^T \underline{v}}$$

↳ coef. matrix of P_V

• reworded:

• $A = [v_1 \dots v_d]$, columns of A L.I. $\Leftrightarrow A^T A$ invertible

• Least-sq. sol:

$$\underline{x} = (A^T A)^{-1} A^T \underline{b}$$

$$A\underline{x} = \underline{A(A^T A)^{-1} A^T \underline{b}} = P_{\text{Col}(A)}(\underline{b})$$

↳ coef. matrix of $P_{\text{Col}(A)}$

12.1 Linear Transformation and coordinates

① Recall

• $[\underline{v}]_B = \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} \leftarrow B\text{-coordinates of } \underline{v}$

• $E_n = \{e_1, \dots, e_n\} \Rightarrow [\underline{v}]_{E_n} = \underline{v} \leftarrow \text{standard coordinates}$

② Matrix that Performs Transformation in B and C coordinates

• What if $T_A: \mathbb{R}^n \rightarrow \mathbb{R}^n$ in non-standard coordinates?

• A $m \times n$ matrix

$B = \{b_1, \dots, b_n\} \subset \mathbb{R}^n$

$C = \{c_1, \dots, c_m\} \subset \mathbb{R}^m$

bases

$$\Rightarrow A_{C \leftarrow B} = [A b_1]_C \dots [A b_n]_C$$

• special case: $A_{E_m \leftarrow E_n} = A$ $I_n = [b_1]_C \dots [b_n]_C = P_{C \leftarrow B}$

• Formula:

$$[c_1 \dots c_m | A b_1 \dots A b_n]_{\text{red}} = [P_C | A P_B]_{\text{red}} = [I_m | A_{C \leftarrow B}]$$

• Application:

$$A_{C \leftarrow B} [\underline{x}]_B = [A \underline{x}]_C$$

$$A_{C \leftarrow B} = P_C^{-1} A P_B$$

$$= \underset{\text{standard} \rightarrow C}{P_{C \leftarrow E_m}} \underset{\text{perform } T_A \text{ in standard}}{A} \underset{B \rightarrow \text{standard}}{P_{E_n \leftarrow B}}$$

12.2 Simplifying Linear Transformations

① Basics

- $m \times n$ A , basis B for $\mathbb{R}^n$, basis C for $\mathbb{R}^m$
- $A_{C \leftarrow B} [x]_B = [Ax]_C$
- $A_{C \leftarrow B} = [Ab_1]_C \dots [Ab_n]_C = P_C^{-1} A P_B = P_{C \leftarrow E_n} A P_{E_n \leftarrow B}$
- Goal: find B and C to make $A_{C \leftarrow B}$ simple

② Rank is Everything

- Rank $A = r \Rightarrow$ can find bases
 $B = \{b_1, \dots, b_r, \underbrace{b_{r+1}, \dots, b_n}_{\text{Null } A}\} \subset \mathbb{R}^n \Rightarrow A_{C \leftarrow B} = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$
 $C = \{c_1, \dots, c_r, c_{r+1}, \dots, c_m\} \subset \mathbb{R}^m$
 $\text{s.t. } Ab_1 = c_1 \dots Ab_r = c_r$
 $Ab_{r+1} = 0 \dots Ab_n = 0$

- Important case: $n=m$, $B=C \leftarrow$ same coords in co- & domain

- M and A similar $\Leftrightarrow M = P^{-1} A P$
 $P = [b_1 \dots b_n]$ some invertible matrix

- $A_{B \leftarrow B} = P_B^{-1} A P_B \Rightarrow A_{B \leftarrow B}$ similar to A
 $\uparrow$ matrices similar to $A =$ matrices performing T_A in other linear coords

③ Simplest form for $A_{B \leftarrow B} = P_B^{-1} A P_B$

- A diagonalizable $\Leftrightarrow A_{B \leftarrow B} = P_B^{-1} A P_B = \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix} \Leftrightarrow Ab_1 = \lambda_1 b_1, \dots, Ab_n = \lambda_n b_n$
 $\uparrow$ Best case short of diagonal $\uparrow$ similar to diagonal matrix $\uparrow$ A scaling B -coords directions $\uparrow$ B eigenbasis

④ Applications

- reflections: $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ diagonalized by $A_{B \leftarrow B} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$
- rotations: $\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \rightarrow$ no b/c no non-zero vector scaled by A
- $A =$ projection matrix $\Rightarrow T_A = P_{V \oplus W}$ & $V = \text{Col } A$
 $W = \text{Null } A$
 $\hookrightarrow B = \{\underbrace{b_1, \dots, b_r}_{\text{Col } A}, \underbrace{b_{r+1}, \dots, b_n}_{\text{Null } A}\} \Rightarrow Ab_1 = b_1 \dots Ab_r = b_r$
 $Ab_{r+1} = 0 \dots Ab_n = 0 \Rightarrow A_{B \leftarrow B} = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} \leftarrow$ diagonal $n \times n$
- A projection $\Leftrightarrow$ Diagonalizable & similar $\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$
- Powers
 - $P^{-1} A P = D \Rightarrow A = P D P^{-1}$
 $\Rightarrow A^k = P \cancel{D P^{-1} P} D P^{-1} \dots P D P^{-1} = P D^k P^{-1} = P \begin{bmatrix} \lambda_1^k & & 0 \\ & \ddots & \\ 0 & & \lambda_n^k \end{bmatrix} P^{-1}$

13.1 Eigenvalues, Eigenvectors, Eigenspaces

① Diagonalizable

- A diagonalizable $\Leftrightarrow A$ similar to a diagonal matrix $\Leftrightarrow$ There exists basis $B \subset \mathbb{R}^n$ s.t.
 $A_{B \leftarrow B} = [Ab_1]_B \dots [Ab_n]_B = P_B^{-1} A P_B$ diagonal
 $\uparrow$ $P^{-1} A P = D$
 $\uparrow$ A scaling in B vector direction
- $A_{B \leftarrow B} = \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix} \Leftrightarrow Ab_1 = \lambda_1 b_1, \dots, Ab_n = \lambda_n b_n$
- Goal: Find non-zero vectors scaled by A

② Eigenvector, Eigenvalue

- A $n \times n$ matrix $\underline{v} \in \mathbb{R}^n =$ eigenvector of A w/ eigenvalue $\lambda \in \mathbb{R}$ if:
 - $\underline{v} \neq 0$
 - $A \underline{v} = \lambda \underline{v}$
 - Note: $\underline{v} \neq 0$ but λ could be zero $\leftarrow \underline{v}$ is 0-eigenvector $\Leftrightarrow \underline{v} \in \text{Null } A$

③ Eigenbasis

- A diagonalizable $\Leftrightarrow$ exists basis for $\mathbb{R}^n$ consisting of eigenvectors of A
 $\uparrow$ $P^{-1} A P$ diagonal $\Leftrightarrow$ columns of $P =$ eigenvectors of A

④ Goal: Find eigenbasis

- (1) Find eigenvalues
 - $\hookrightarrow \det(A - \lambda I) = 0 \Leftrightarrow \text{Null}(A - \lambda I_n) \neq 0 \Leftrightarrow (A - \lambda I_n) \underline{v} = 0$ for some $\underline{v} \neq 0$
 - $\hookrightarrow$ Characteristic polynomial: $\chi_A(t) = \det(A - t I_n)$
 - λ an eigenvalue $\Leftrightarrow \chi_A(\lambda) = 0$
 - $A = \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix} \Rightarrow \chi_A(t) = (\lambda_1 - t) \dots (\lambda_n - t) \Rightarrow \lambda_1, \dots, \lambda_n$ are eigenvalues
- (2) Find eigenvectors
 - $\hookrightarrow A \underline{v} = \lambda \underline{v} \Leftrightarrow (A - \lambda I_n) \underline{v} = 0 \Leftrightarrow \underline{v} \in \text{Null}(A - \lambda I_n)$
 - $\hookrightarrow \lambda$ -eigenspace: subset of all λ -eigenvectors and $0 \leftarrow \lambda = 0, E_0 = \text{Null } A$
 $E_\lambda = \text{Null}(A - \lambda I_n)$
 $\uparrow \lambda$ eigenvalue $\Leftrightarrow \dim E_\lambda \geq 1$

① Diagonalizing

① Diagonalizing

- diagonalizing $A \Leftrightarrow$ Find P w/ $P^{-1}AP$ diagonal $\Leftrightarrow$ finding eigenbasis of A

② Strategy to Diagonal Matrix A

- (1) Calculate & completely factor $\chi_A(t) = \det(A - tI_n)$
- (2) Find bases for each eigenspace $E_\lambda = \text{Null}(A - \lambda I_n)$
- (3) Take union & hope we have basis

③ Algebraic Multiplicity

- **Algebraic Multiplicity**: $\chi_A(t) = (\lambda_1 - t)^{m_{\lambda_1}} \dots (\lambda_k - t)^{m_{\lambda_k}} \Rightarrow m_{\lambda_i}$
- $m_{\lambda_1} + \dots + m_{\lambda_k} = n \leftarrow \deg \chi_A(t) = n$
- A upper triangular $\Rightarrow m_{\lambda} =$ number of time λ appears on main diagonal
- $1 \leq \dim E_{\lambda} \leq m_{\lambda} \leftarrow$ if λ an eigenvalue, can find at most m_{λ} L.I. λ -eigenvectors
- * unions of L.I. eigenvectors in different eigenspaces are L.I.

④ Is A Diagonalizable?

- Criterion: A diagonalizable $\Leftrightarrow \dim E_\lambda = m_\lambda$
- A has n distinct eigenvalues $\Rightarrow A$ diagonalizable
 - $\uparrow$ $m_\lambda = 1$ and $1 \leq \dim E_\lambda \leq m_\lambda = 1 \Rightarrow \dim E_\lambda = m_\lambda = 1$
- not distinct eigenvalues
 - $\hookrightarrow$ Alg mult $m_\lambda > 1 \rightarrow$ need to calculate $\dim E_\lambda = \text{Null}(A - \lambda I)$
 - $\rightarrow$ if $\dim E_\lambda < m_\lambda$
 - $\rightarrow$ not diagonalizable
 - $\rightarrow$ # free columns = # dim

14.1 Symmetric Matrices and the Spectral Theorem

① Recall Diagonalizable

- A diagonalizable $\Leftrightarrow P^{-1}AP = \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix} \Leftrightarrow$ columns of P form an eigenbasis B
- Diagonalizing A = find P w/ $P^{-1}AP$ diagonal = Find eigenbasis of A
- $\Rightarrow$ study diagonalization for symmetric matrices

② symmetric Matrix

- A symmetric $\Leftrightarrow A^T = A$
- Special properties
 - $\underline{u} \cdot (A\underline{v}) = (A^T \underline{u}) \cdot \underline{v} \Rightarrow$ If A symmetric: $\underline{u} \cdot (A\underline{v}) = (A\underline{u}) \cdot \underline{v}$
- A symmetric $\Rightarrow$ All eigenvalues are real
- A symmetric $\Rightarrow$ A preserves $V \Rightarrow A$ preserves $V^\perp$
 - $\Rightarrow$ A preserves $V \neq \{0\} \Rightarrow V$ contains an eigenvector of A

- Note: A orthogonally diagonalizable $\Leftrightarrow P^T A P = \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix} \Leftrightarrow$ columns of P form an orthogonal eigenbasis
- ↑
diagonal matrices
orthog. diagonalized by standard basis
- Spectral Theorem

③ Spectral Theorem

- A real symmetric $\Leftrightarrow$ A orthogonally diagonalizable
 - $\{b_1, \dots, b_n\}$ eigenvectors of A
 - $\downarrow$
 - A preserves V
 - $V^\perp$ contains an eigenvector b_{n+1} of A
 - $\uparrow$
 - A preserves $V^\perp$
- * $d=n \Leftarrow \{b_1, \dots, b_n\}$ is orthonormal eigenbasis
- A symmetric $\Rightarrow$ Different eigenspaces are orthogonal

④ Orthogonally diagonalize

- compute $\lambda_A(t)$ & factor
- for each λ s.t. $\chi_A(\lambda) = 0$, compute basis for $E_\lambda = \text{Null}(A - \lambda I_n)$
- for any i s.t. $\dim E_{\lambda_i} > 1$, run G.S. of basis E_{λ_i} to get orthogonal basis
- normalize
- take union
- $P^T A P \rightarrow$ orthogonally diagonalize any real symmetric matrix

14.2 Singular Value Decomposition

• Present linear transformation $T_A: \mathbb{R}^n \rightarrow \mathbb{R}^m$ in simplest way

① Recall Rank is Everything

• given A $m \times n$ matrix, there exists orthonormal bases:

$$B = \{b_1, \dots, b_n\} \subset \mathbb{R}^n \text{ and } C = \{c_1, \dots, c_m\} \subset \mathbb{R}^m \text{ s.t.}$$

$$Ab_1 = c_1, \dots, Ab_r = c_r, Ab_{r+1} = 0, \dots, Ab_n = 0$$

$$\Rightarrow A_{CB} = P_C^{-1} A P_B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \text{ rank} = r$$

② Singular Value Decomposition

• restrict ourselves to orthonormal bases $\rightarrow$ orthonormally diagonalize

$\hookrightarrow$ eigen as V

• given A $m \times n$ matrix, there exists orthonormal bases

$$V = \{v_1, \dots, v_n\} \subset \mathbb{R}^n \text{ and } U = \{u_1, \dots, u_m\} \subset \mathbb{R}^m \text{ s.t.}$$

$$Av_1 = \sigma_1 u_1, \dots, Av_r = \sigma_r u_r, Av_{r+1} = 0, \dots, Av_n = 0$$

$$\Rightarrow A_{UV} = U^T A V = \Sigma = \begin{bmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_r & 0 \\ 0 & 0 & 0 \end{bmatrix} \text{ rank} = r$$

$$U = [u_1 \dots u_m] \quad V = [v_1 \dots v_n] \quad \sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0$$

$\hookrightarrow$ singular values

$B V^T$
 $\downarrow$
SVD

$$\Rightarrow A = U \Sigma V^T \leftarrow \text{SVD of } A$$

• $A^T A$ orthogonally diagonalized by V

• Spectral Theorem $\Rightarrow$ SVD

$\uparrow$
symmetric matrices
are orthogonally diagonalizable

$\leftarrow$ All matrices
admit a SVD

③ SVD Algorithm

• orthogonally diagonalize $A^T A$

• reorder so eigenvalues decreasing $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n \geq 0$

$$\hookrightarrow \text{define } V = [v_1 \dots v_n]$$

• $\sigma_i = \sqrt{\lambda_i}$. we can define Σ

$$\text{define } u_1 = \frac{1}{\sigma_1} A v_1, \dots, u_r = \frac{1}{\sigma_r} A v_r \Rightarrow \{u_1, \dots, u_r\} \text{ ONB for Col } A$$

• extend $\{u_1, \dots, u_r\}$ to ONB for $\mathbb{R}^m$

$\hookrightarrow$ find a basis for $(\text{Col } A)^\perp = \text{null } A^T$

$\hookrightarrow$ Apply G.S. to get orthogonal basis for $(\text{Col } A)^\perp$

$\hookrightarrow$ normalize to get $\{u_{r+1}, \dots, u_m\}$

• define U & write SVD $\rightarrow A = U \Sigma V^T$

$$A = U \Sigma V^T$$